Assignment-02: Comparative Study and Practical Analysis of Generative AI Models

Autoencoder (AE) | Variational Autoencoder (VAE) | DCGAN | Stable Diffusion

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1. Introduction and Objective

This report compares four generative models, studied in two pairs. **Part A (CO3)** covers image reconstruction with an Autoencoder and a Variational Autoencoder. **Part B (CO4)** covers image generation with a Deep Convolutional GAN and a pretrained diffusion model (Stable Diffusion). The objective is to explain how each model works, run ready-made Google Colab implementations, and analyse the outputs for blurring, distortion, loss of detail, realism and limitations. No model was developed from scratch; only short demonstration training runs were used for AE, VAE and DCGAN.

2. Comparative Study of the Four Models

Table 1 summarises the six required criteria. The last row uses what we observed in our own runs (Section 3).

Criterion	Autoencoder	VAE	DCGAN	Stable Diffusion
Working principle	Squeezes an image into a small code and rebuilds it; trained only to minimise reconstruction error (MSE in our run).	Encodes each image as a Gaussian (mean, variance), samples a code with the reparameterisation trick and decodes it. Loss = reconstruction + KL divergence to N(0, I).	A generator turns noise into images while a discriminator separates real from fake; the two are trained against each other.	Starts from random noise in a compressed latent space and removes it step by step, steered by the text prompt; trained to reverse gradual noising [4,5].
Architecture and components	Conv encoder, 16-value bottleneck, transposed-conv decoder (0.18 M parameters).	Same decoder, but the encoder ends in two heads (mean, log-variance) plus a sampling layer (about 0.53 M parameters).	Generator: dense layer, transposed convolutions, BatchNorm. Discriminator: strided convolutions, LeakyReLU, no pooling [3] (generator 2.33 M parameters).	CLIP-based text encoder, U-Net denoiser with cross-attention, VAE decoder from latent to pixels, noise scheduler [5].
Advantages	Simple and fast to train; good for compression, denoising and anomaly detection.	Stable training; smooth latent space, so sampling and interpolation work.	Sharper images than a VAE; generates in a single forward pass.	Highest realism and detail; text control; diverse outputs; stable training.
Limitations	Unstructured latent space, so it cannot generate new images; loses detail through the bottleneck.	Blurrier than GAN or diffusion; the KL term trades sharpness for latent structure.	Unstable training, mode collapse, no direct control over the output, sensitive to hyper-parameters.	Slow (many denoising steps), heavy memory and compute; artifacts such as hands and text; inherits dataset bias.
Applications	Compression, denoising, dimensionality reduction, anomaly detection.	Sample generation, latent interpolation, representation learning, anomaly detection.	Image synthesis, data augmentation, face and style generation.	Text-to-image, inpainting, image editing, design and concept art.
Output differences (our runs)	Close copies of simple shapes, smoothed texture (mean MSE 0.0108). Random codes decode to unrecognisable blobs.	Slightly blurrier than AE (mean MSE 0.0145). Random codes decode to recognisable garments.	Sharper new 28x28 samples, but some are malformed; no prompt control.	High-resolution colour output that follows the prompt; slowest per image.

Table 1: Comparison of AE, VAE, DCGAN and Stable Diffusion on the six criteria.

3. Practical Analysis

Setup. All runs used Google Colab with a T4 GPU and TensorFlow 2.21 (notebooks 1-3). AE, VAE and DCGAN were trained on Fashion-MNIST (28x28 grayscale clothing images), chosen because textures and logos make blur easy to see. AE and VAE share the same eight test images (indices 0-7, which serve as the input images), the same 16-value bottleneck and the same decoder layout. Stable Diffusion v1.5 (stable-diffusion-v1-5) was used pretrained and run inside Colab on a Tesla T4 with the Hugging Face diffusers library (512x512 output, 30 steps, guidance 7.5, seed 42, three fixed prompts), so nothing was trained.

Model	Configuration	Parameters	Time	Mean MSE (8 images)
Autoencoder	10 epochs, MSE loss, batch 128	178,001	49.7 s (training)	0.0108
VAE	20 epochs, BCE + KL loss, batch 128	about 0.53 M	87.1 s (training)	0.0145
DCGAN	20 epochs, batch 128, noise size 100	2.33 M (generator)	326.7 s (training)	not applicable
Stable Diffusion	pretrained SD v1.5, 30 steps, 512x512, T4 GPU	1,066 M (U-Net 859.5 M)	6.1 to 9.3 s per image	not applicable

Table 2: Configuration and measured values taken from the executed notebooks. Parameter counts are from each notebook's model summary; the VAE total adds the encoder (424,480) to the decoder (108,993, computed from its layer sizes).

Activity A: Image reconstruction (CO3)

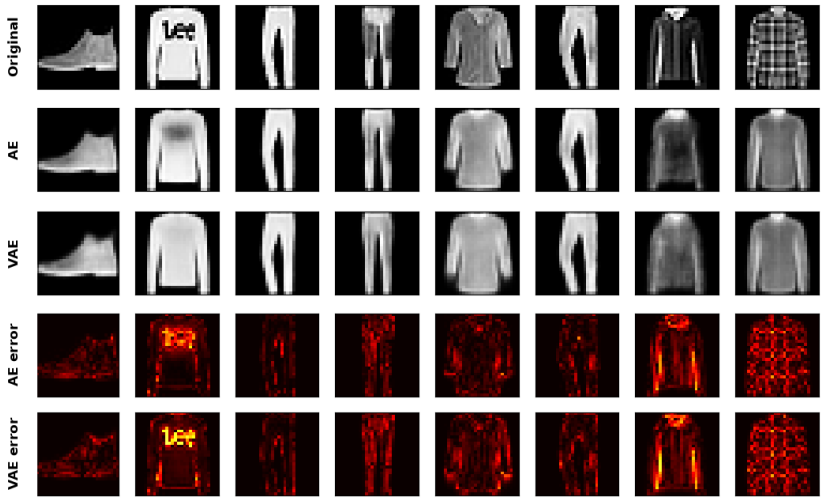


Figure 1: Input images (Original) with AE and VAE reconstructions and their absolute-error maps (brighter = larger error). Screenshots A1-A3 from notebooks 1 and 2.

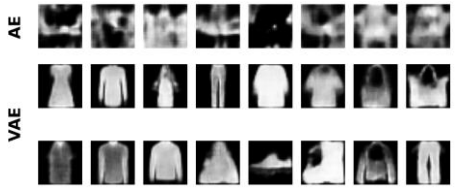


Figure 2: Images decoded from random latent vectors (AE above, VAE below).

Image	AE	VAE
Ankle boot	0.0032	0.0036
"Lee" sweater	0.0207	0.0324
Plaid shirt	0.0225	0.0231
Dark jacket	0.0215	0.0284
Mean of 8	0.0108	0.0145

Table 3: Per-image reconstruction MSE (lower is better).

- **Blurring and loss of detail, not geometric distortion, is the main defect.** Both models keep silhouettes correct (trousers, ankle boot, MSE below 0.01) but lose fine texture. The error maps are brightest on logos, plaid, zips and collars.
- **VAE is blurrier and less accurate than AE on every one of the eight images** (mean MSE 0.0145 vs 0.0108, about 34% higher), even though the VAE was trained for 20 epochs against 10 for the AE. The gap is therefore not explained by under-training. On the "Lee" sweater the AE leaves a smudged dark patch where the logo was, while the VAE removes it completely and returns a plain sweater. VAE outputs also drift towards a generic prototype of their class (several tops look alike).
- **For the plaid shirt both models fail almost equally** (0.0225 vs 0.0231): the pattern cannot be stored in 16 numbers, so both return a flat grey shirt.
- **The VAE pays for this blur with a usable latent space.** Random vectors through the AE decoder give unrecognisable shapes (Figure 2), whereas the VAE produces recognisable dresses, tops, trousers, a sneaker and an ankle boot, although edges are soft and a few are malformed. The VAE's KL term settled at about 15 nats per image, which limits how much detail the code can carry and is consistent with the blur we see.

- **Fairness note.** Our notebooks used different training lengths (10 vs 20 epochs) and different losses (MSE vs BCE + KL). Averages are from only eight images, so the ranking should be read as indicative, not statistical.

Activity B: Image generation (CO4)



Figure 3: DCGAN, 64 samples after 20 epochs from fixed noise (screenshot B1).

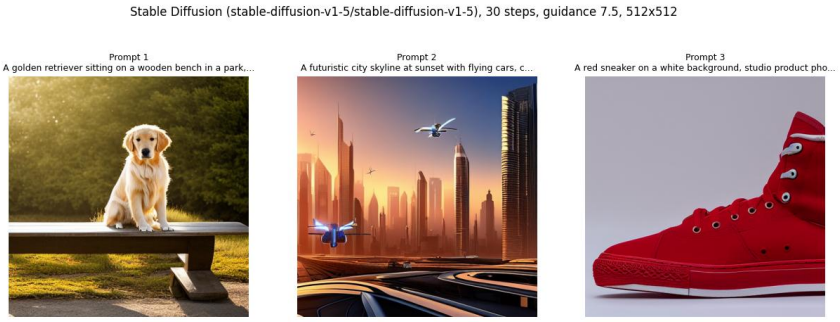


Figure 4: Stable Diffusion v1.5 output for the three fixed prompts (golden retriever on a bench; futuristic city at sunset; red sneaker, studio photo), 30 steps, guidance 7.5, seed 42, 512x512 (screenshot B2_SD_all).

- **DCGAN.** After 20 epochs (326.7 s) the grid shows tops, trousers, sneakers and boots in clear variety, so we see no obvious mode collapse in these 64 samples. Edges and fabric texture are crisper than the VAE samples in Figure 2, but several samples are malformed (merged or half-formed shapes, speckled backgrounds). At epoch 5 most samples were still shapeless blobs, so quality improved steadily with training. The losses flattened near G = 0.77 and D = 1.32 from about epoch 10; these are close to the theoretical equilibrium values (ln 2 = 0.69 and 2 ln 2 = 1.39), but a flat loss does not measure image quality, so the grid itself had to be inspected. Output is limited to 28x28 grayscale with no control over which item appears.
- **Stable Diffusion.** The pretrained Stable Diffusion v1.5 ran in Colab on a Tesla T4 and produced 512x512 colour images from text. Its U-Net alone has 859.5 M parameters (CLIP text encoder 123.1 M, VAE 83.7 M) against 2.33 M for our DCGAN generator, and each image needs 30 denoising steps against one generator pass, taking 9.3 s for prompt 1 and 6.1 s each for prompts 2 and 3. In Figure 4 the dog has realistic fur and soft backlight, and the city has a coherent sunset gradient with detailed towers. Secondary details are weaker: the “flying cars” look like drones, and the sneaker is cropped, on a pale grey rather than white background. Time grows with steps: for prompt 1, 5, 15 and 30 steps took 1.2 s, 3.2 s and 6.3 s, and the 5-step image was softer.
- **Comparison.** DCGAN is cheap and sharp but uncontrolled and small-scale; diffusion gives prompt control and far higher fidelity at a much higher cost per image. The two are not tested on the same data, so this is a comparison of model families as run in these notebooks, not a controlled benchmark.

4. AI Usage and Reflection

AI tool used: Claude (Anthropic), for planning the work pack, drafting the research notes and Colab notebooks, debugging, and drafting this report from the executed notebooks. **Verification process:** (1) every number in Section 3 was read from the executed notebook outputs, not from the AI’s memory; (2) the code was executed, which exposed a real bug; (3) AI-written pack text was compared with the notebooks, which exposed two mismatches; (4) theory statements are tied to the references below and were checked against them **Manmath: confirm after ticking the source-verification table**. Important prompts are summarised in Table 4.

No.	Prompt (summary)	Issue found and fix	Verified with
1	Read the brief and tracker, then prepare Manmath’s pack: research notes, comparison table, four Colab notebooks, run guide.	Draft needed human review before use.	Assignment rubric (brief, Section 5).

2	Build AE and VAE notebooks in Keras with identical inputs for a fair comparison.	A helper named _losses clashed with a Keras internal attribute and crashed training; renamed to _compute_losses.	Executed in Colab on Fashion-MNIST.
3	Build a DCGAN notebook and a Stable Diffusion notebook that uses a hosted API.	The hosted-API Stable Diffusion notebook needed a token and could not be run, so it was replaced by a no-key version that runs SD v1.5 in Colab with diffusers.	Both notebooks executed in Colab (DCGAN; SD v1.5 on a Tesla T4).
4	Write research notes and a comparison table for the four models.	AI-written content must be checked against sources.	Papers [1]-[5]; ticks pending.
5	Read the three executed notebooks, the tracker and the brief, then draft the report (Krishna's part).	The pack said MNIST and "same epochs", but the notebooks use Fashion-MNIST and 10 vs 20 epochs; report follows the notebooks. The first Notebook 4 had no outputs; Figure 4 comes from the re-run no-key version.	Notebook outputs and code cells.

Table 4: Important prompts, issues found and how each was verified.

Key learning outcomes.

- Reconstruction accuracy and generative ability pull in opposite directions: the AE reconstructs better, but only the VAE's regularised latent space can generate.
- Blur in a VAE comes from its objective (pixel-wise loss plus KL penalty), so it is expected behaviour rather than a bug.
- GAN loss curves do not show image quality; samples must be inspected, and variety checked for mode collapse.
- Diffusion buys realism and text control with many sequential steps and large models; AI-generated documents can contradict the code, so outputs must be checked against what was actually run.

5. Contributions of Group Members

- **Manmath (Research and Code Lead):** researched the four models, drafted the comparison table and source list, prepared and debugged the four notebooks (including the VAE fix), maintained the prompts log and reviewed the technical content.
- **Vishal (Execution and Evidence Lead):** set up Colab with a T4 GPU, ran the notebooks, captured the named screenshots, recorded runtimes and filled the observation table.
- **Krishna (Documentation Lead):** wrote this report and the PPT, analysed the executed outputs, built the figures and tables, checked the page limit and references, and assembled the submission.

References

[1] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," ICLR 2014, arXiv:1312.6114.

[2] I. Goodfellow et al., "Generative Adversarial Nets," NeurIPS 2014, arXiv:1406.2661.

[3] A. Radford, L. Metz and S. Chintala, "Unsupervised Representation Learning with Deep Convolutional GANs," ICLR 2016, arXiv:1511.06434.

[4] J. Ho, A. Jain and P. Abbeel, "Denoising Diffusion Probabilistic Models," NeurIPS 2020, arXiv:2006.11239.

[5] R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR 2022, arXiv:2112.10752.

[6] G. E. Hinton and R. R. Salakhutdinov, "Reducing the Dimensionality of Data with Neural Networks," Science 313, 2006; I. Goodfellow, Y. Bengio and A. Courville, Deep Learning, MIT Press, 2016, Ch. 14.

[7] H. Xiao, K. Rasul and R. Vollgraf, "Fashion-MNIST," arXiv:1708.07747, 2017; Hugging Face Diffusers documentation; Keras and TensorFlow example tutorials (autoencoder, VAE, DCGAN).